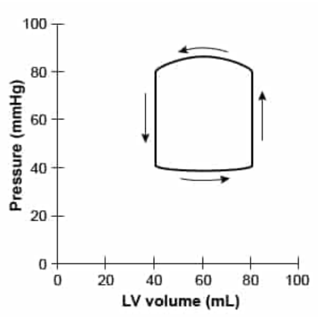
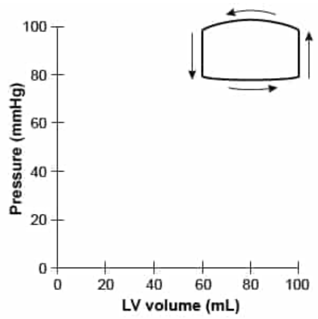
Researchers measured the pressure and volume changes in the hearts of volunteers. Which of the following graphs represent the greatest amount of work done by the heart?

○ A.



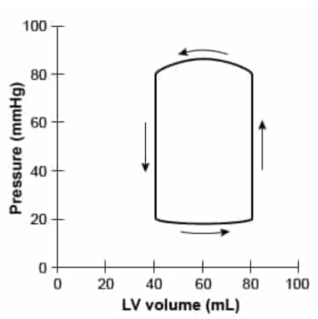
[3%]

○ B.



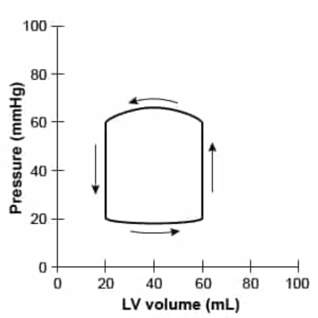
[14%]

✔ ◉ C.



[78%]

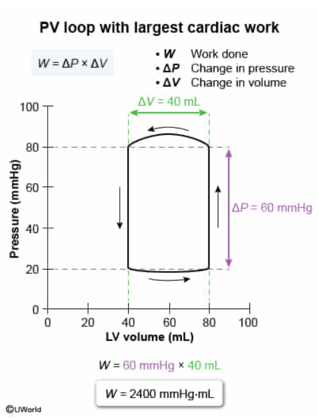
○ D.



[3%]

Correct
78% answered correctly

Explanation:



The work W done equals the product of the force F applied and the distance d:

$$W = \text{Fd}$$

A cardiac PV loop plots the pressure *P* and **volume** *V* of the left ventricle during the cardiac cycle. Hence, the PV loop **area** equals the product of *P* and *V*:

$$\text{PV Loop Area} = P \times V$$

Furthermore, pressure equals the ratio of *F* and the area *A* over which it is applied:

$$P = \frac{F}{A}$$

Combining the two above equations yields:

$$\text{PV Loop Area} = \frac{F}{A}V = F\frac{V}{A}$$

The ratio of *V* in m^3 and *A* in m^2 results in a value with units of meters, corresponding to distance. Consequently, this equation simplifies to:

$$\text{PV Loop Area} = \text{Fd} = W$$

Therefore, the area of a PV loop equals the *W* done by the heart.

In this question, *W* equals the product of the changes in pressure ΔP and volume ΔV, because the PV loops do not start at zero *P* or *V*:

$$W = \Delta P \times \Delta V$$

Calculating *W* for the diagram shown in Choice C gives:

$$W = (80 \text{ mmHg} - 20 \text{ mmHg}) \times (80 \text{ mL} - 40 \text{ mL})$$

$$W = 60 \text{ mmHg} \times 40 \text{ mL} = 2{,}400 \text{ mmHg} \cdot \text{mL}$$

The nonstandard units of *W*, mmHg·mL, are sufficient to determine which graph represents the greatest *W*. Calculating *W* for the diagrams in Choices A, B, and D gives 1,600 mmHg·mL, 800 mmHg·mL, and 1,600 mmHg·mL, respectively. Therefore, the PV loop shown in Choice C represents the greatest amount of work by the heart.

(Choices A and D) These PV loops represent 1,600 mmHg·mL of cardiac work each, which is less than 2,400 mmHg·mL of work.

(Choice B) This PV loop has the greatest values of pressure and volume, but the work done by the heart depends on the changes in pressure and volume, not just their maximum values.

Educational objective:
A cardiac PV loop records the pressure and volume in the left ventricle during a complete cardiac cycle. The area enclosed by the PV loop represents the work done by the heart.

Time spent:0 sec
Subject:
Physics

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System:
Fluids